

Math 1 • E25 re-replace review

Only E25 after ejecting the f^n composition replacement. For checking - not a full paper.

E25 • REPLACE AGAIN

prior replacement used f^n composition/iteration (Maths 2 / MM); still out of Math 1 spec

The graph of a quadratic polynomial $y = P(x)$ is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ to give the graph of $y = Q(x)$.

The polynomial $Q(x)$ has roots at $x = 2$ and $x = 6$.

Given that the graph of $y = P(x)$ touches the x -axis exactly once, what is the y -intercept of the graph of $y = P(x)$?

A $-\frac{49}{2}$

B -1

C $-\frac{1}{2}$

D $\frac{1}{2}$

E 6

F $\frac{49}{2}$